Washington X. Quevedo

XR + AI Engineer · Unity, VR/AR & Applied ML

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Summary

XR engineer with 10 years building VR/AR training, simulation and interactive experiences in Unity, now combining them with applied AI: computer vision, on-device inference and LLM-based agents. Published researcher in virtual reality, with 30+ peer-reviewed papers, 640 citations and an h-index of 14, plus a registered software IP and a national research award. Shipped XR and game work for a social VR platform (Sansar, via BairesDev), an AR company (Merge) and a pharmaceutical sales force, and currently builds the AI module of a Stanford University research platform. Founder of Inmersoft (2018), with 200+ delivered projects.

Experience

Laboratorios Bagó del Ecuador — Digital Solutions & AI Developer

10/2020 – present · Quito (pharmaceutical company)

- Delivered a group VR session on 30 synchronized Meta Quest headsets: one Unity app (360° guided tour, meditation, interactive game) controlled from a Windows panel over a local network, with per-headset battery status; all 360° video shot and edited in-house.
- Developed vSpace, the medical-visit platform used by the sales force, with 20 embedded mini-games reaching 400 users and 20,000 monthly interactions.
- Built the MKT Attack event web app (live schedule, QR-based uploads, live ranking), a corporate metaverse and 12 augmented-reality filters for brand campaigns.
- Built an AI-assisted platform for the product managers' yearly creative event (MKT Attack): schedules the sessions, shows who presents and the time left, and gives teams product information as context for ideas; 8 product managers, 12 campaigns.
- Built TrackingApp, a logistics platform (Laravel 12, Inertia, React) covered by 246+ automated tests.

Sansar (Linden Lab) — Game Developer (contract via BairesDev)

2022 · Remote

- Built avatar interaction mechanics and a dynamic animation system for a social VR platform.
- Optimized shaders and materials, reducing memory usage by ~40%.

Stanford University — Software & AI Consultant (Buscapez project)

07/2026 – present · Remote · contracted through Inmersoft as an official Stanford supplier

- Built the species-identification module for a tuna-tagging research platform: a YOLO detector trained in PyTorch (mAP50 0.96 for detection on validation), exported to ExecuTorch for offline on-device inference on iOS and Android and to ONNX for a FastAPI service (49 ms per image).
- Diagnosed input distortion in the mobile runtime and fixed it with letterboxing; species classification keeps improving as the team validates more real photos.
- Delivered the web platform, live in production; the mobile app is in development.

Inmersoft — Founder & Principal Engineer

09/2018 – present · Quito

- 200+ delivered projects. VR/AR applications for engineering, medicine and entertainment: industrial training, rehabilitation and immersive process environments.
- Digital twins of drones and robots for control-algorithm testing; Unity ML-Agents.
- Hardware integration in XR: Leap Motion, Emotiv EEG, haptics, Arduino and IoT devices.

Merge — AR Research Engineer

2020

- Led the migration research from Vuforia to Unity AR Foundation, unifying ARKit/ARCore object recognition and validating cross-platform 3D detection.

Universidad de las Fuerzas Armadas ESPE — VR/AR Research Analyst

2016 – 2018

- Built VR/AR systems for robotics simulation, industrial training and rehabilitation that led to most of the publications listed below.

Earlier roles

- **Arlift** — Platform Engineer, contract (10/2025 – present): QHSE compliance platform, Laravel API + React SPA, 308 tests.
- **WebCreek** — Unity 3D Developer (2019 – 2020).
- **Inter-American Development Bank** — Platform Developer, GoHelp platform (2016 – 2017).

Research & Recognition

- 30+ peer-reviewed publications on VR, AR and robotics (Springer: AVR, ISVC; ACM ICETC; CISTI) · 640 citations · h-index 14 · i10-index 21 (Google Scholar, Oct 2026). Peer reviewer.
- Most cited: Unity3D-MatLab simulator in real time for robotics applications (88); Augmented reality as a new marketing strategy (55); Virtual reality on e-Tourism (54); Virtual reality system for training in automotive mechanics (first author, 50).
- Registered software IP: Virtual 2D Laser Sensor, a C# LIDAR simulation connected to MATLAB, Ecuadorian Intellectual Property Institute, QUI-052058 (2017).
- First place, SENESCYT National Awards (Galardones Nacionales), 2016.

Selected XR Projects

SkyNest — multiplayer VR combat sandbox (in development)

Unity 6, Meta Quest, Netcode for GameObjects, Laravel + React

- Physics-based hands, melee and ranged weapons, armor with durability and grenades; PvP duels on a self-hosted game server; passwordless account flow entirely inside VR; admin panel and backend deployed to a VPS with GitHub Actions.

RayNeo Power — HUD for RayNeo Air 4 Pro glasses (in development)

Swift, macOS, iOS

- Heads-up display for AR glasses: macOS host with 3DoF head tracking, and an iPhone app that renders its own window on the glasses (clock, YouTube with audio that keeps playing while the phone is locked) with the phone as remote control.

Earlier XR work (behance.net/wxquevedo)

- **VR Runner:** maze game on a static treadmill with HTC Vive, jump detection and muscle and heart-rate sensors.
- **DJI Matrice 600 Pro VR simulator:** virtual GPS, IMU and altimeter plus two 5-DOF arms, linked to a MATLAB UAV model to test autonomous control and teleoperation (HTC Vive, Leap Motion).
- **Unity3D–MATLAB real-time robotics simulator:** basis of the most-cited paper (88 citations).
- Also: robot teleoperation in VR, P&ID diagram to immersive 3D plant, virtual automotive assembly, reconstruction of historic buildings, AR filters and apps (Spark AR, Lens Studio, AR Foundation, Vuforia, Three.js, A-Frame).

Skills

XR & games	Unity, C#, VR (Meta Quest, PC VR), AR (ARKit/ARCore, AR Foundation, Vuforia), shaders and optimization, Unity ML-Agents, haptics and sensor integration
AI / ML	Python, PyTorch, YOLO, ONNX Runtime, ExecuTorch, computer vision, on-device inference, ComfyUI
LLMs & agents	Anthropic Claude API (primary), OpenAI and Google Gemini APIs, Claude Code, MCP, tool calling, local LLM serving (llama.cpp)
Web & mobile	React, TypeScript, React Native / Expo, Swift, Laravel, FastAPI, PostgreSQL, MySQL, Redis, Linux, Git

Education

B.Eng. in Electronics and Instrumentation Engineering — Universidad de las Fuerzas Armadas ESPE, Ecuador (2009 – 2015)

Languages

Spanish (native) · English (professional working proficiency)